LEWICE Training

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Main Input File

- **Emphasis on key inputs**
- **Data ranges**
- **Warning/Error messages**
- **User Tips**

This first training session will familiarize the user with the different fields in the input files, their meaning, and appropriate ranges where applicable.

Some warning and error messages as well as user tips are also covered. This training is not comprehensive and the user may have to refer to the User Manual for further details.

All input files are plain text (ASCII) files. The extensions .INP and .XYD were used in the example cases for identification purposes only.

LEW20 Namelist

&LEW20	Units	Validated Range
ITIMFL = 1	None	1
TSTOP = 300.	Seconds	$0 < t \leq 2700$
IBOD = 1	None	1
IFLO = 5	None	Calculated
DSMN = 4.D-4	None	$2.D\text{-}4 \leq d \leq 8.D\text{-}4$
NPL = 24	None	24
RHOP = 1000.	Kg/m^3	1000
IGRID = 0	None	0
IDEICE = 0	None	0
&END		

This is an example of the LEW20 namelist in the main input file.

All of the data in the main input file is entered through namelists.

LEW20 Namelist

- **ITIMFL selects automated time stepping**
  - **= 0, runs user selected time steps**
  - **= 1 (recommended), sets minimum number of time steps**
- **DSMN selects point spacing**
  - **(Total Wrap Distance)/DSMN = # of points on baseline geometry**
  - **Panel geometry then determined from baseline geometry**
- **IGRID selects flow solver**
  - **= 0 (recommended), potential flow**
  - **= 1, reads grid and flow solutions from file**
    - » **Single time step only**
- **IDEICE selects 1D deicing**
  - **Does not affect ice accretion**
  - **Provides first order estimate of heat loads**

Minimum number of time steps (for ITIMFL = 1) is found from the equation

N = LWC * V * Time /(Chord * Ice Density * 0.01)

The user can choose to run more time steps (set with IFLO) than calculated by this equation.

The standard procedure is to use this equation to define the number of time steps for longer (>10 min.) cases and to use a time step of 1 minute for shorter (<10 min.) runs

DSMN validated range is 0.0004 to 0.0008 (dimensionless). One DSMN value is input for each body input so that the flap can have a higher point density than the main element (for example).

Grid file format is PLOT3D binary format with IBLANK on. LEWICE can read and use multi-block structured grids. Both standard and Chimera grids were used in testing of this option.

More detailed deicer/anti-icier predictions can be performed with ANTICE and LEWICE/Thermal

Warning Messages

- **Notifies user when value is outside validated range**
- **Notifies user if feature has not been validated**
- **Notifies user if input is invalid**

ITIMFL = 0 or 1; other values will be changed so ITIMFL = 1

TSTOP must be > 0 ; no data past 45 minutes

IBOD must be from 1 to 5

IFLOW must be > 0

DSMN must be > 0; warning if over 8.D-4

NPL must be ≥ 10; over 50 consider unnecessarily excessive

RHOP = 1000. for water

IGRID = 0 or 1; other values changes so that IGRID = 0

IDEICE = 0 or 1; other values changes so that IDEICE = 0

DIST Namelist

&DIST
FLWC = 0.05, 0.1, 0.2, 0.3, 0.2, 0.1, 0.05
DPD = 6.2, 10.4, 14.2, 20.0, 27.4, 34.8, 44.4
&END

- **FLWC is a fraction (no units)**
- **DPD = Drop Size in Microns**

This is an example of the DIST namelist in the main input file.

FLWC = Volume Fractional Liquid Water Content for each drop size (values must add up to 1)

DPD = drop size for each bin size in microns (10 bins max)

MVD is found from the distribution so it is not input

LEWICE 2.0 automatically applies correction for very small drops so the correction is not an input variable in this version.

Warning Messages

- **Notifies user when value is outside validated range**
- **Notifies user if input is invalid**

The sum of the FLWC values = 1.

A warning will be printed to the screen for exceedence conditions due to the limited number of data points available.

Drop sizes ranged from 15 µm to 270 µm in the validation database.

ICE1 Namelist

&ICE1	Units	Validated Range
CHORD = 0.9144	m	$0.35 \le c \le 2$
AOA = 0.0	Degrees	$-4 \le \alpha \le 7$
VINF = 89.5	m/s	$56 \le V \le 146$
LWC = 0.34	g/m^3	$0.31 \le LWC \le 1.8$
TINF = 269.	Kelvin	$241 \le T \le 270$
PINF = 100000.0	N/m^2	100000
RH = 100.0	Percent	100
&END		

Angle of attack is the angle of the flow with respect to the geometry coordinants input.

High (>6˚) angles and/or high velocities can cause flow separation which may not be modeled well.

Temperature and pressure input are static, not total, values.

Experimental data ranges:

Chord : 14 in to 78 in

AOA : -4 to +7

Velocity: 56 m/s to 146 m/s

Mach No.: 0.17 to 0.45

Reynolds No.: $2*10^6$ to $13*10^6$

LWC: 0.31 to 1.8 g/m^3

MVD: 15 µm to 270 µm

Temperature (static): -25 ˚F to 27 ˚F

Temperature (total): -15 ˚F to 33 ˚F

ICE1 Namelist

- **For multi-element cases, chord is a reference length**
- **Corrections are made to AOA values when using Potential Flow**
- **Temperature is static temperature**
- **Nominal values for pressure and relative humidity were used in validation tests**

The body geometry and DSMN values are input as dimensionless values. So long as (x/c) * c gives x in meters, any reference length can be used for chord (theoretically).

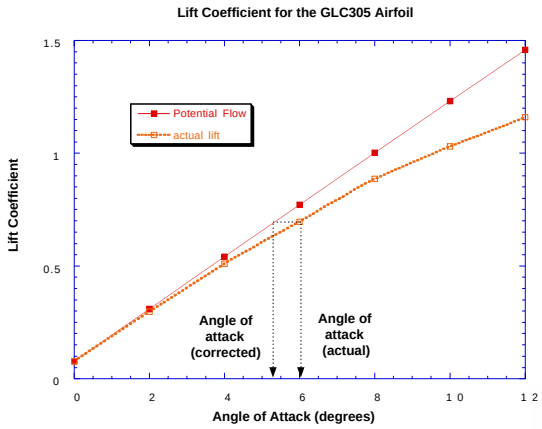
For the validation test, the AOA value input was less than the “true” angle of attack to correct for the overprediction of lift made by the potential flow code.

Make sure the variables are input in the proper units.

Since pressure (altitude) and relative humidity are secondary factors to the ice accretion, nominal values can be used for the validation tests since the true values were not recorded.

Warnings are printed if input values are outside validation database.

Angle of Attack Correction



This plot illustrates the correction applied to angle of attack when running validation cases for LEWICE. The curved line represents the lift curve of the clean airfoil while the straight line represents the lift predicted by LEWICE. The correction uses the angle of attack which in LEWICE will produce the same lift coefficient as would be expected on the clean airfoil. Especially for large ice shapes generated at high angles of attack, this correction may not be enough.

For multi-element cases, the flap deflection has also been changed in order to match the lift coefficient.

LPRNT Namelist

```
&LPRNT
FPRT = 1
HPRT = 1
BPRT = 1
EPRT = 0
MPRT = 0
TPRT = 0
IDBF = 0
&END
```


This namelist controls how much of the data is written to the various output files. The flags are all set so that a value of zero results in no output, a value of one results in some output and a value of two results in all output for that file (applicable only to the larger output files).

For example, TPRT=0 means that individual particle trajectories will not be written, TPRT=1 means that the trajectories used to find collection efficiency (i.e. those which hit the body) are written and TPRT=2 means that all particle trajectories are written out. This last option can result in large output files.

Body Geometry

- **Each body is input from a separate file**
- **Each line consists of an x,y coordinate pair**
- **Geometry is input as dimensionless values**

Good geometry input

1.0000	0.000000
0.9950	-0.000520
0.9900	-0.000990
0.9800	-0.001820
0.9700	-0.002560
0.9600	-0.003290
0.9500	-0.004030
0.9250	-0.005870
0.9000	-0.007710
.	
.	
.	

Bad geometry input

xheader	yheader
78.000	0.0000
77.610	-0.040560
77.220	-0.077220
76.440	-0.14196
75.660	-0.19968
74.880	-0.25662
74.100	-0.31434
72.150	-0.45786
.	
.	
.	

Geometry Errors

- **Geometry not closed**
- **Typing errors for coordinates**
- **Insufficient resolution**
  - **50 to 200 points per body in validation tests**
- **Blank lines in file**
  - **Not applicable to all systems**
- **Clockwise coordinates**
- **Values not dimensionless**
- **Headers in file**

Most input file errors result from a poorly defined airfoil geometry. Some of the more common errors are listed above.

LEWICE will tend to smooth the input geometry (as well as ice shapes). If your input geometry has sharp corners which you wish to preserve, make certain to provide good resolution to that region in your input file.

Interactive Input and Output

- **Prompts user for name of main input file and geometry input file.**
- **Prompts user to confirm warning messages (if any)**
- **Errors (and corrective actions) are described in user manual.**

For each case, LEWICE will prompt the user for the name of the main input file containing the namelist input and the name of a geometry input file. For a multi-body case, additional prompts will be issued for the user to enter the name of the file(s) containing the additional body(s).

If LEWICE detects a problem with any of the input files, it will issue warning and/or error messages which describe the problem. For warning messages, LEWICE will continue to run after the user confirms the warning. Error messages will cause it to stop running. Every warning and error message is listed in the user manual with suggested corrective action.

All output to the screen is also output to the file “junk.dat”

Output Files

- **ICE1.DAT**
  - **Clean and iced geometry for first body**
  - **Output for other bodies in ICE2.DAT, ICE3.DAT, etc.**
  - **Data for each successive time step follows the previous time step output**
  - **Output coordinates in inches for easier comparison with experimental data**
- **FLOW.DAT**
  - **Potential flow solution**
- **BETA.DAT**
  - **Collection efficiency output**
  - **Body geometry also output for reference**
- **HTC.DAT**
  - **Heat transfer coefficients**

LEWICE can output several files, each of which may contain useful information. Every file can not be covered in this presentation.

All output files are plain text (ASCII) files and have a .DAT extension.

Output files can easily be read into spreadsheet programs or other packages for post-processing.

Each column of a file has a header identifying then variable, including units.

Output Files (cont.)

- **IMP.DAT**
  - **Impingement limit output**
    - » **Wrap measured from leading edge and from stag. Pt.**
- **FINAL1.DAT**
  - **Contains only final ice shape**
    - » **Format compatible with feature measurement utility (THICK)**
  - **FINAL2.DAT, FINA3.DAT for other bodies**
- **MISC.DAT**
  - **Individual particle trajectories**
  - **Lift coefficient**
  - **Inertia parameters**
  - **Stagnation point**
  - **Echo of main input file**

IMP.DAT contains the locations of the uppermost and lowermost trajectory hits. The location where beta=0 in the BETA.DAT file may not be the same due to panel resolution.

Output Files (cont.)

- **JUNK.DAT**
  - **Debug information**
  - **Listing of any error or warning messages**
- **TRAJ1.DAT**
  - **Particle trajectories for first body**
  - **TRAJ2.DAT, TRAJ3.DAT, etc. for other bodies**
- **PRES.DAT**
  - **Flow variables used in boundary layer calculation**
- **Other files described in user manual**

Depending on the flags set in the LPRT namelist, LEWICE can output many files, some of which may not be useful for a given situation. The user is encouraged to turn off flags which will write out information which is not needed.

This presentation can not cover all of the output files and the specific information in them. Please refer to the user manual for a complete description.

Utility Programs

- **THICK**
  - **Calculates ice thickness and other parameters for validation tests**
- **CONVERT**
  - **Converts a version 1.6 input file to a version 2.0 input file**
- **TOASCII**
  - **Converts PLOT3D files from BINARY format to ASCII format**
- **TOBINARY**
  - **Converts PLOT3D files from ASCII format to BINARY format**

Five utility programs are included with LEWICE as aids for the user.

The THICK program was used on both LEWICE and experimental ice shapes to extract measurements for comparison such as horn height, horn angle, and icing limits,

CONVERT will convert input files from version 1.6 format to version 2.0 format, but it may be just as easy to create the version 2.0 input files from scratch.

EXPAND will increase/decrease the number of points in a geometry file by using linear interpolation.

TOASCII and TOBINARY can be used to manipulate PLOT3D files for use in LEWICE or for transfer to other systems. As binary formats for different platforms are not the same, the binary grid and flow solution files enclosed must be converted to be used on other platforms.

Discussion

- **Desired input options**
- **Output options**
- **Graphical user interface**
- **Interactive plotting of output**

What other features would be useful for LEWICE?

Are the input and output formats easy to use?

Would a GUI (Graphical User Interface) be more beneficial?

What about interactive plotting as seen in the demo?